Function Machines

Year 4 · **Lesson Worksheet**

Worked Examples

One-Step Machine: A function machine applies the rule ×4 to its input. What is the output when the input is 7?

Answer: 7 × 4 = 28.

Two-Step (Chained) Machine: A function machine: input → +3 → ×2 → output. What is the output when the input is 5?

Answer: 5 + 3 = 8; 8 × 2 = 16.

Working Backwards: A function machine: input → ×3 → −7 → output. The output is 11. Find the input.

Answer: Reverse the steps: 11 + 7 = 18; 18 ÷ 3 = 6. The input was 6.

Function Basics (15 questions)

1. A function machine applies the rule ×5 to its input. What is the output when the input is 6?
2. A function machine applies the rule +9 to its input. What is the output when the input is 14?
3. A function machine applies the rule −6 to its input. What is the output when the input is 20?
4. A function machine applies the rule ÷4 to its input. What is the output when the input is 32? A) 6 B) 7 C) 8 D) 9 E) 10
5. A two-step function machine: input → +2 → ×3 → output. What is the output when the input is 4?
6. A two-step function machine: input → ×2 → +5 → output. What is the output when the input is 9? A) 18 B) 20 C) 21 D) 23 E) 25
7. A function machine applies the rule ×6 to its input. If the output is 42, what was the input?
8. A function machine applies the rule +15 to its input. If the output is 31, what was the input? A) 14 B) 15 C) 16 D) 17 E) 18
9. A function machine applies the rule −8 to its input. If the output is 5, what was the input?
10. A function machine applies the rule ÷5 to its input. If the output is 9, what was the input?
11. Select ALL that apply: a function machine applies the rule ×3 to each input. Which of these input→output pairs are correct? A) 4→12 B) 5→15 C) 6→20 D) 7→21 E) 8→22
12. A two-step function machine: input → ×4 → −3 → output. What is the output when the input is 5?
13. A two-step function machine: input → +7 → ÷2 → output. What is the output when the input is 9?
14. A function machine takes an input, doubles it, then adds 6. What is the output when the input is 10? A) 20 B) 22 C) 24 D) 26 E) 28
15. A function machine applies the rule −10 to its input. What is the output when the input is 10?

Working Forwards & Backwards (15 questions)

1. A function machine: input → ×3 → +8 → output. What is the output when the input is 11?

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2. A function machine: input → −5 → ×4 → output. What is the output when the input is 12? A) 24 B) 26 C) 28 D) 30 E) 32

3. A function machine: input → ÷3 → −2 → output. The input is 18. Find the output.

4. A machine takes an input, adds 4, then multiplies by 3. If the output is 33, what was the input? A) 5 B) 6 C) 7 D) 8 E) 9

5. A function machine: input → ×2 → −9 → output. The output is 15. Find the input.

6. A function machine turns 5 into 19 and 8 into 28 using the same two-step rule "×a, then +b". Find the rule (give both a and b).

7. A function machine: input → ×4 → output. What is the output when the input is 0.5?

8. Priya puts a number into a machine. The machine triples the number, then adds 5. The output is 29. What number did Priya put in? A) 6 B) 7 C) 8 D) 9 E) 10

9. A function machine: input → +7 → output. What is the output when the input is −3?

10. Select ALL that apply: a function machine applies the rule "subtract 4, then multiply by 2" to inputs. Which input→output pairs are correct? A) 6→4 B) 9→11 C) 10→12 D) 12→14 E) 15→22

11. A function machine: input → ×5 → output. What is the output when the input is 2/5?

12. A machine doubles a number and then subtracts 4. If the result is 10, what was the original number?

13. A function machine: input → ×0.5 → output. What is the output when the input is 13? A) 5.5 B) 6 C) 6.5 D) 7 E) 7.5

14. A function machine: input → ÷4 → +6 → output. What is the output when the input is −8?

15. A function machine takes a number, adds 1/4 to it, then doubles the result. What is the output when the input is 3/4?

Multi-Step Challenge (15 questions)

1. Nadia puts a number into a machine. It multiplies the number by 3, then adds 5. What is the output when she puts in 7?

2. A machine's rule is 'take away 6, then multiply by 4'. If the output is 20, what was the input?

3. A rule machine does '×3, then add b' to any input. Put in 5 and you get 19. What is b? A) 2 B) 3 C) 4 D) 5

4. Start with a number, divide it by 4, then add 3. The result is 9. What number did you start with?

5. A machine's rule is '×5, then subtract 9'. What do you get if you put in 6?

6. A puzzle machine works out 'double it, then take away 5'. Select ALL of the pairs below that this machine could produce: A) 4→3 B) 6→7 C) 8→11 D) 10→16 E) 12→19

7. Start with a number. Add 15, then divide by 5. If the result is 6, what number did you start with? A) 10 B) 12 C) 15 D) 18 E) 20

8. Take a number, multiply it by 6, then subtract 14. The answer is 4. What was the number?

9. A machine adds 11 to a number, then doubles the result. What is the output if you start with 4?

10. Halve a number, then subtract 4. Starting with 10, what do you get?

11. A puzzle: double a number, add 6, and you get 20. What was the number? A) 5 B) 6 C) 7 D) 8 E) 9

12. At a party, hiring a room costs £3, plus £4 for every guest. For 6 guests, what is the total hire cost?

13. A number has 5 taken away from it, and the answer is then shared equally into 3 parts, each part being 4. What was the original number?

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14. Multiply a number by 5, then add 5. Starting with 5, what is the result? A) 25 B) 28 C) 30 D) 32 E) 35

15. Multiply a number by 3, then subtract 4. What do you get if you start with 6?

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